OpenLESS: Automated Landing Error Scoring System

Subject Data: OpenCapData_71 (Jump Landing Trials) | **Algorithm Reference:** Turner et al. (2024), *medRxiv* DOI: 10.1101/2024.11.28.24318160

Report Generated: October 13, 2026 | **Processing:** OpenCap Camera 60 Hz, 4th-order Butterworth Low-Pass 12 Hz

Trial 1: Stepoff-Leaning

Total LESS Score: 1 / 19

Rating: EXCELLENT / LOW RISK

Soft landing with active sagittal dissipation (deep knee & hip flexion, forward trunk lean). Symmetric forefoot initial ground contact.

Trial 2: Stepoff-Upright

Total LESS Score: 14 / 19

Rating: POOR / HIGH RISK

Stiff, erect landing with restricted flexion, upright rigid spine, asymmetric heel-strike, lateral hip shift, and dynamic knee valgus collapse.

Itemized Landing Error Scoring System (LESS) Criteria

#	LESS Item	Operational Criteria (Turner et al., 2024)	Stepoff-Leaning	Stepoff-Upright
1	**Knee Flexion (IC)**	< 30° knee flexion at initial contact	**1** (L: 27.0°, R: 19.8°)	**1** (L: 23.8°, R: 9.8°)
2	**Hip Flexion (IC)**	< 30° hip flexion at initial contact	**0** (L: 38.2°, R: 33.1°)	**1** (L: 24.8°, R: 17.6°)
3	**Trunk Flexion (IC)**	< 10° trunk flexion (> -10° lumbar extension)	**0** (-18.2°)	**1** (-6.3°)
4	**Ankle Plantarflexion**	Heel contact prior to or simultaneous with toes	**0** (Toe-first contact)	**1** (L: Heel-first, R: Toe)
5	**Asymmetrical Timing**	One foot lands > 34 ms before the other	**0** (0.0 ms diff)	**0** (16.7 ms diff)
6	**Asymm. Heel/Toe Strike**	One foot lands heel-first, other toe-first	**0** (Both toe-first)	**1** (L heel vs R toe)
7	**Lateral Trunk Flexion**	> 5° lateral lumbar bending to either side	**0** (1.8°)	**0** (-1.7°)
8	**Medial Knee Pos. (IC)**	Hip rot > 8° & add > -1°, or hip add > 1°	**0** (rot: -6.3°, add: -3.7°)	**0** (rot: -8.3°, add: -2.4°)
9	**Wide Stance Width**	> 12° hip abduction at IC (< -12° adduction)	**0** (L: -3.7°, R: -4.6°)	**0** (L: -2.4°, R: -6.3°)
10	**Narrow Stance Width**	> 0° hip adduction at initial contact	**0** (L: -3.7°, R: -4.6°)	**0** (L: -2.4°, R: -6.3°)
11	**Foot Inward Rotation**	> 10° subtalar angle inward (IR)	**0** (L: -2.9°, R: -2.5°)	**0** (L: -8.8°, R: -2.5°)
12	**Foot Outward Rotation**	> 15° subtalar angle outward (< -15°)	**0** (L: -2.9°, R: -2.5°)	**0** (L: -8.8°, R: -2.5°)
13	**Knee Flexion (Max)**	< 65° maximum knee flexion in stance	**0** (L: 80.1°, R: 81.4°)	**1** (L: 54.8°, R: 63.3°)
14	**Hip Flexion (Max)**	< 45° maximum hip flexion at peak knee flexion	**0** (L: 79.2°, R: 80.2°)	**1** (L: 39.3°, R: 43.7°)
15	**Trunk Flexion (Max)**	< 10° trunk displacement between IC and Max	**0** (Δ: 27.0°)	**1** (Δ: 0.1°)
16	**Medial Knee Pos. (Max)**	Hip rot > 8° & add > -1°, or hip add > 1° at Max	**0** (L add: -4.2°, R: -7.3°)	**1** (L add: 1.1°, R: -9.6°)
17	**Asymmetric Loading**	> 5° difference in max hip adduction (shift)	**0** (Diff: 3.0°)	**1** (Diff: 10.8°)
18	**Sagittal Displacement**	0 = Soft (>55° knee, >73° hip), 1 = Avg, 2 = Stiff	**0** (0 (Soft landing))	**2** (2 (Stiff landing))
19	**Overall Impression**	0 = Excellent, 1 = Average, 2 = Poor	**0** (0 (Excellent))	**2** (2 (Poor))
Σ	**TOTAL LESS SCORE**	**Sum of all 19 Movement Error Items**	**1 / 19**	**14 / 19**

Biomechanical Trajectories & Kinematic Comparison

Continuous 3D joint kinematics during the stance phase (initial contact to toe-off) extracted from OpenCap markerless motion capture and analyzed through the OpenLESS automated pipeline.

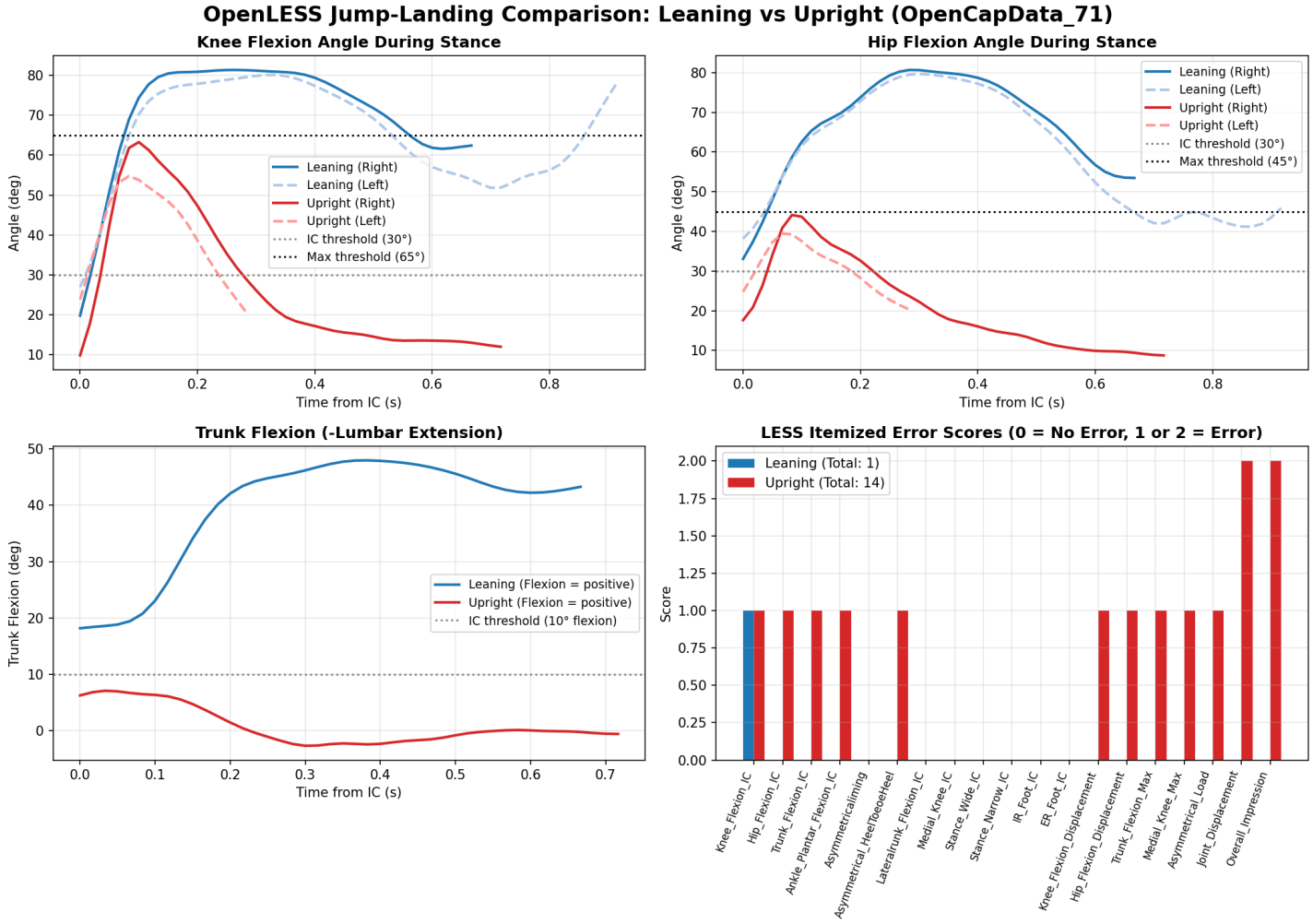


Figure 1: Joint kinematics during landing stance comparing Stepoff-Leaning (blue) and Stepoff-Upright (red), with clinical thresholds and itemized LESS error score distribution.

Sagittal Plane Energy Dissipation: In *Stepoff-Leaning*, the jumper proactively flexes the hips (peak 80.2°) and knees (peak 81.4°) accompanied by a pronounced forward trunk lean (45.2°), effectively distributing vertical ground reaction forces across the lower limb musculature. In contrast, *Stepoff-Upright* exhibits a rigid landing strategy with limited knee flexion (54.8° on left), limited hip flexion (39.3°), and an upright rigid spine (displacement Δ = 0.1°), which substantially elevates peak ground impact and joint contact stresses transmitted directly through the tibiofemoral joint.

Frontal Plane Alignment & Valgus Collapse: In *Stepoff-Upright*, as the left limb reaches peak landing depth, the hip moves into positive adduction (+1.1°), producing dynamic knee valgus collapse. Furthermore, an asymmetric hip adduction discrepancy of 10.8° between right and left limbs indicates pronounced lateral trunk/hip shifting, a primary clinical predictor of non-contact anterior cruciate ligament (ACL) injury.

Jump-Landing Event Detection & Quality Control

Automated detection of Initial Ground Contact (IC) and Toe-Off (TO) from 4th-order low-pass filtered (12 Hz) vertical coordinates of big toe and heel markers, validating stance phase bracketing.

Trial ID	Limb	Toe Minima (s)	Heel Minima (s)	Contact Type	IC Time (s)	TO Time (s)	Stance Duration (s)
Stepoff-Leaning	Left	1.800	1.900	Toe-first	1.800	2.717	0.917
Stepoff-Leaning	Right	1.800	1.850	Toe-first	1.800	2.467	0.667
Stepoff-Upright	Left	1.117	1.067	Heel-first	1.067	1.350	0.283
Stepoff-Upright	Right	1.050	1.067	Toe-first	1.050	1.767	0.717

Marker Vertical Trajectories & Ground Contact Verification

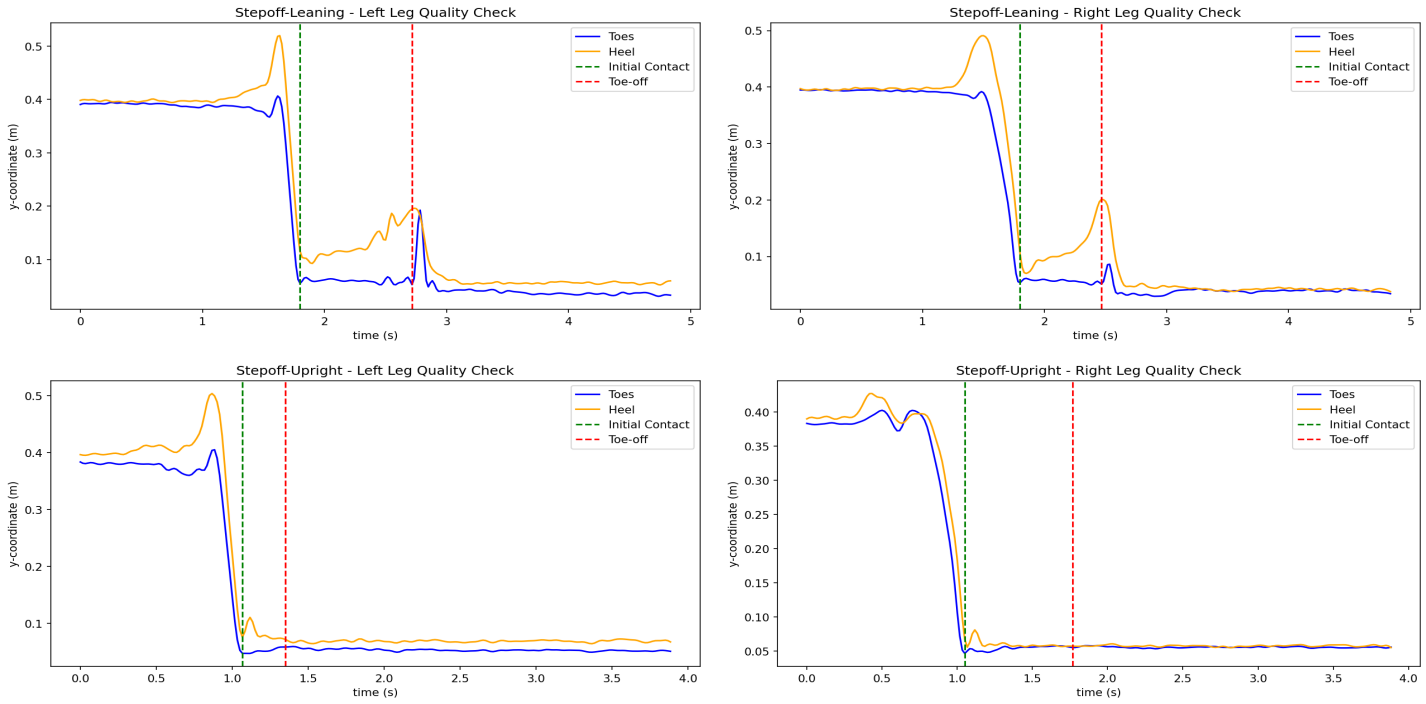


Figure 2: Quality control verification plots displaying filtered heel (orange) and toe (blue) vertical trajectories. Dashed green line marks initial contact (IC); dashed red line marks toe-off (TO).

Data Artifacts Generated:

- Comprehensive Scores: OpenCapData_71/less_scores_final.csv (68 parameters)
- Summary LESS Scores: OpenCapData_71/less_short_final.csv (19 criteria + Total)
- Clipped/Filtered Stance Kinematics: OpenCapData_71/OpenLESS_Results/*_kin_*_filtered.csv
- Python Pipeline: run_openless_scoring.py (Turner et al., 2024 implementation)